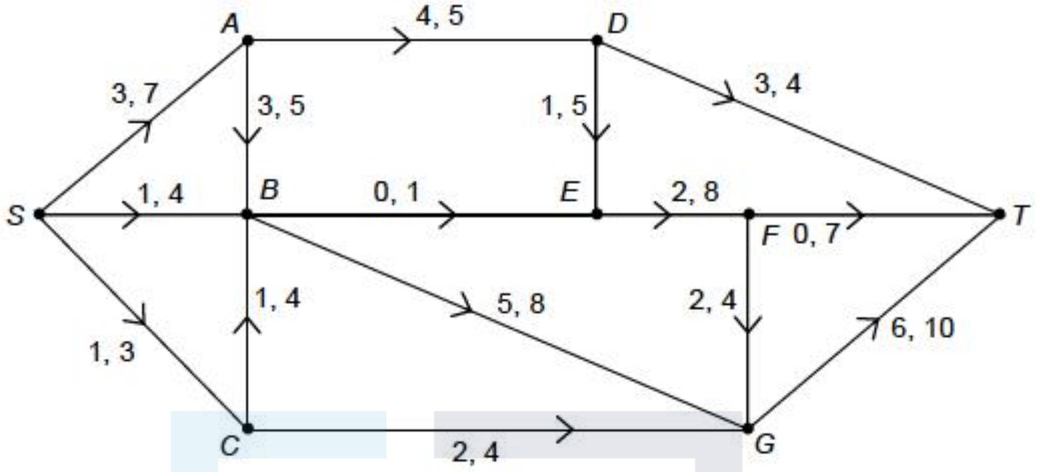
Q1.

The network shows a system of pipes, where *S* is the source and *T* is the sink.

The lower and upper capacities, in litres per second, of each pipe are shown on each arc.



(a) There is a feasible flow from *S* to *T*.

(i) Explain why arc *AD* must be at its lower capacity.

(1)

(ii) Explain why arc *BE* must be at its upper capacity.

(1)

(b) Explain why a flow of 11 litres per second through the network is impossible.

(1)

(c) The network in **Figure 1** shows a second system of pipes, where *S* is the source and *T* is the sink.

The lower and upper capacities, in litres per second, of each pipe are shown on each edge.

Figure 1

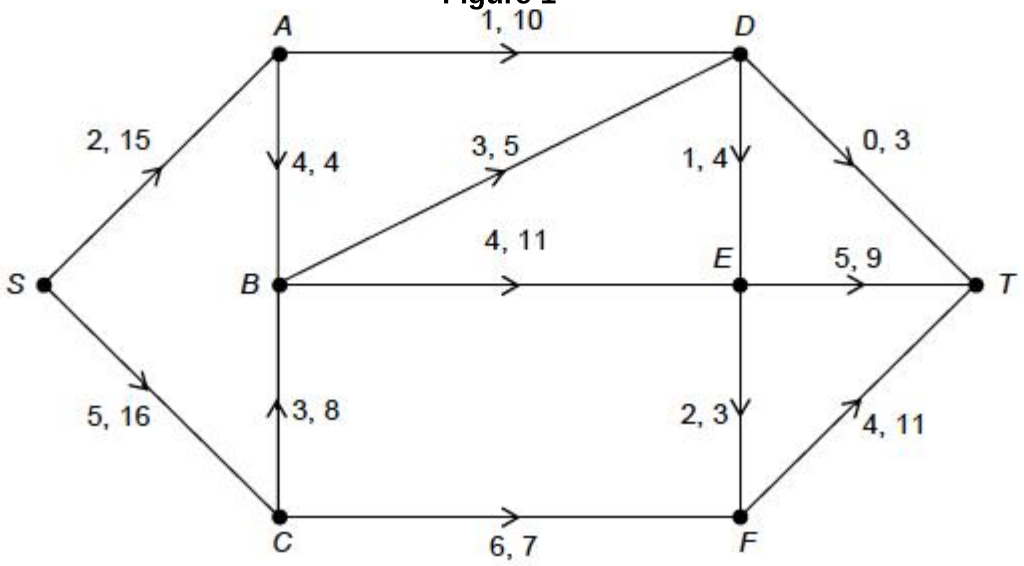
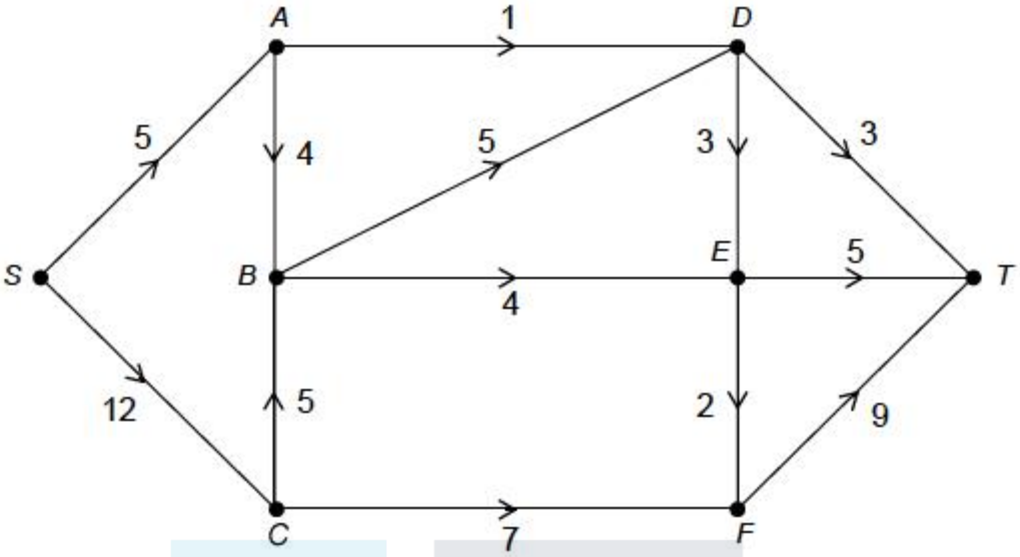


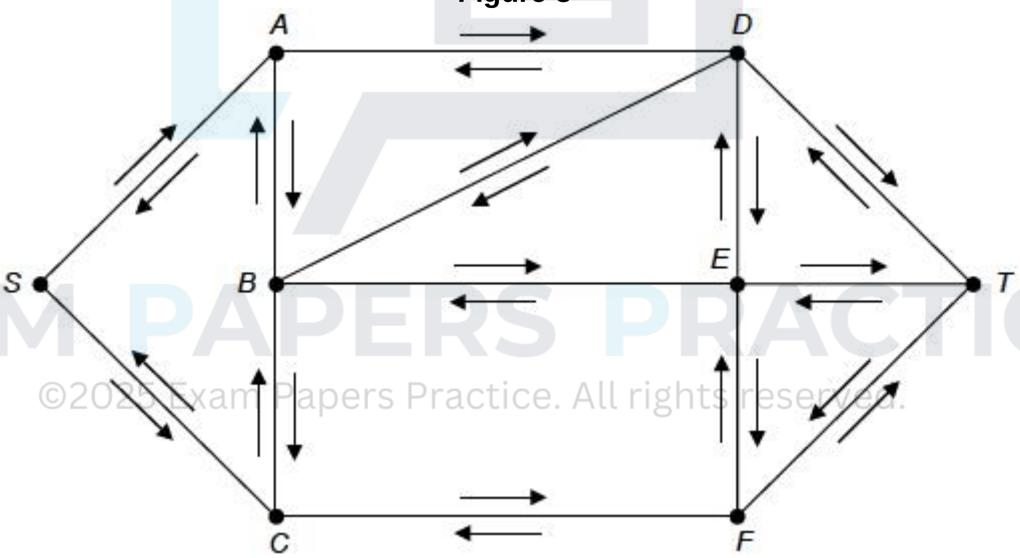
Figure 2 shows a feasible flow of 17 litres per second through the system of pipes.

Figure 2



(i) Using **Figures 1** and **2**, indicate on **Figure 3** potential increases and decreases in the flow along each arc.

Figure 3



(2)

(ii) Use flow augmentation on **Figure 3** to find the maximum flow from S to T.

You should indicate any flow augmenting paths clearly in the table below and modify the potential increases and decreases of the flow on **Figure 3**.

(1)

Augmenting Path	Flow

(iii) Prove the flow found in part **(c)(ii)** is maximum.

(1)

(iv) Due to maintenance work, the flow through node *E* is restricted to 9 litres per second.

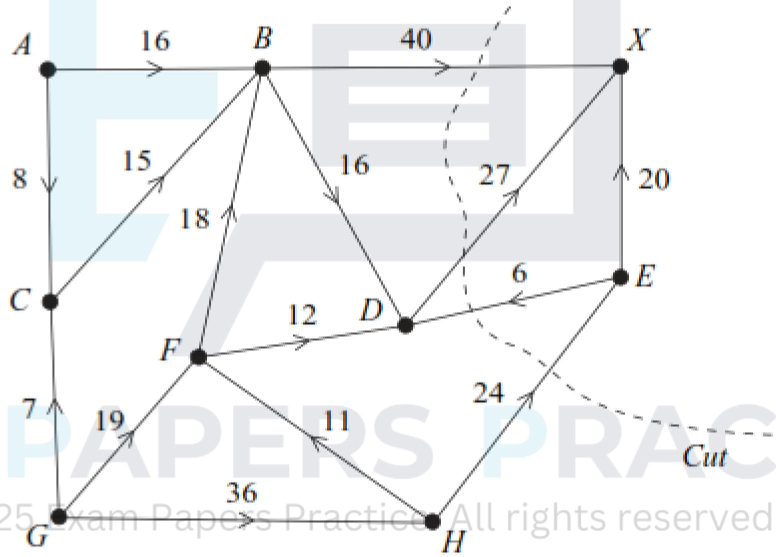
Interpret the impact of this restriction on the maximum flow through the system of pipes.

(2)

(Total 11 marks)

Q2.

The network shows the evacuation routes along corridors in a college, from two teaching areas to the exit, in case of a fire alarm sounding.



The two teaching areas are at A and G and the exit is at X.

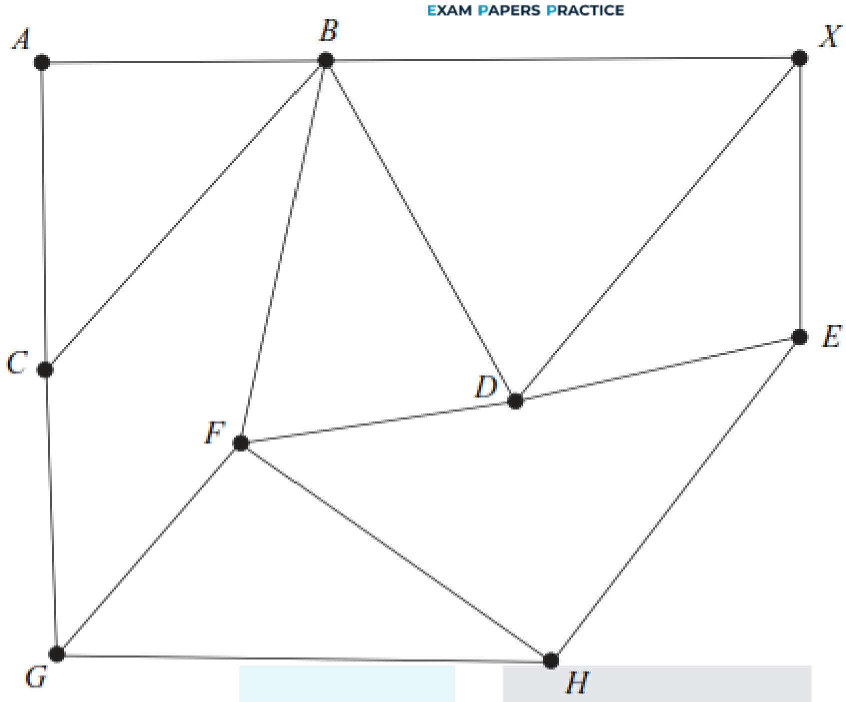
The number on each edge represents the maximum number of people that can travel along a particular corridor in one minute.

(a) Find the value of the cut shown on the diagram.

(1)

(b) Find the maximum flow along each of the routes $ABDX$, $GFBX$ and $GHEX$ and enter their values in the table on **Figure 1**.

Figure 1



Route	Flow
$ABDX$	
$GFBX$	
$GHEX$	

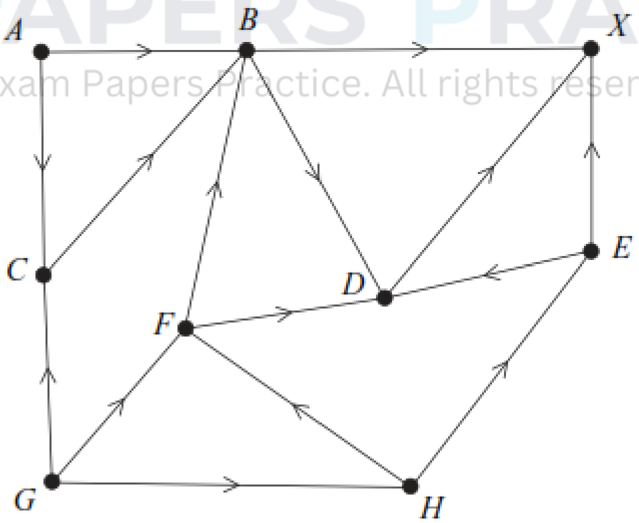
(3)

(c) (i) Taking your answers to part (b) as the initial flow, use the labelling procedure on **Figure 1** to find the maximum flow through the network. You should indicate any flow augmenting routes in the table and modify the potential increases and decreases of the flow on the network.

(5)

(ii) State the value of the maximum flow, and, on **Figure 2**, illustrate a possible flow along each edge corresponding to this maximum flow.

Figure 2



Maximum flow is ______________________ people per minute.

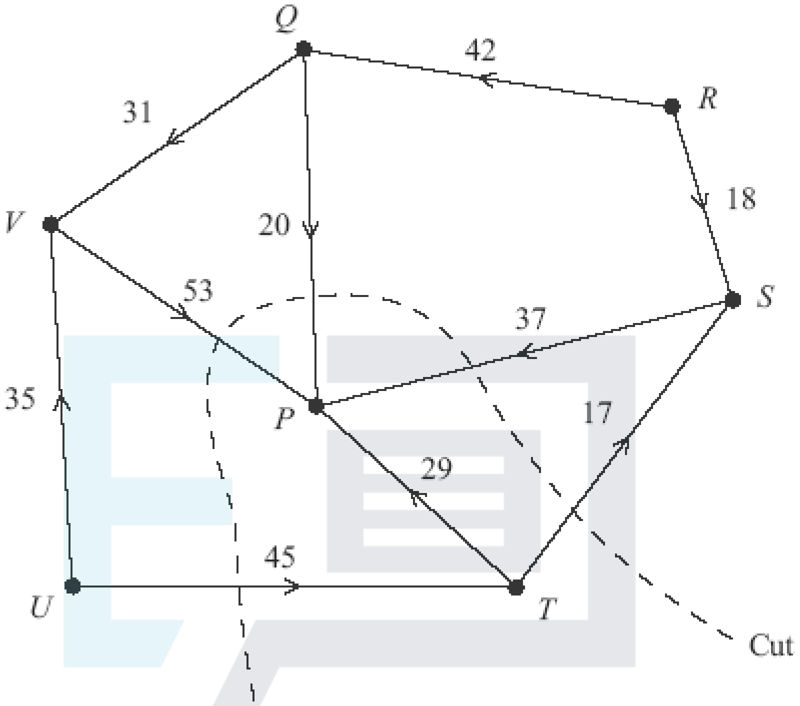
(2)

(d) During one particular fire drill, there is an obstruction allowing no more than 45 people per minute to pass through vertex B. State the maximum number of people that can move through the network per minute during this fire drill.

(2)
(Total 13 marks)

Q3.

The network shows the routes along corridors from two arrival gates to the passport control area, P, in a small airport. The number on each edge represents the maximum number of passengers that can travel along a particular corridor in one minute.



(a) State the vertices that represent the arrival gates.

(1)

(b) Find the value of the cut shown on the network.

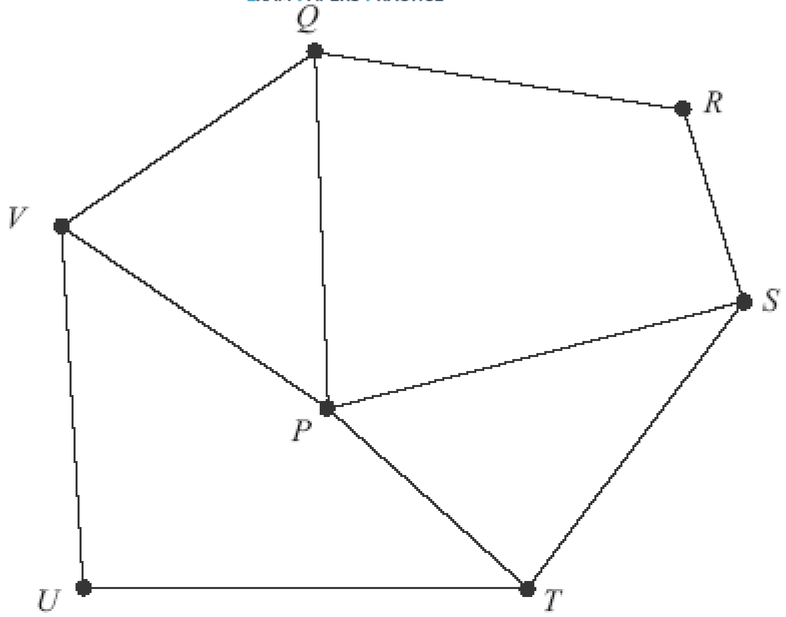
(1)

(c) State the maximum flow along each of the routes $UTSP$ and $RQVP$.

(2)

(d) (i) Taking your answers to part (c) as the initial flow, use the labelling procedure on **Figure 1** to find the maximum flow through the network. You should indicate any flow augmenting paths in the table and modify the potential increases and decreases of the flow on the network.

Figure 1



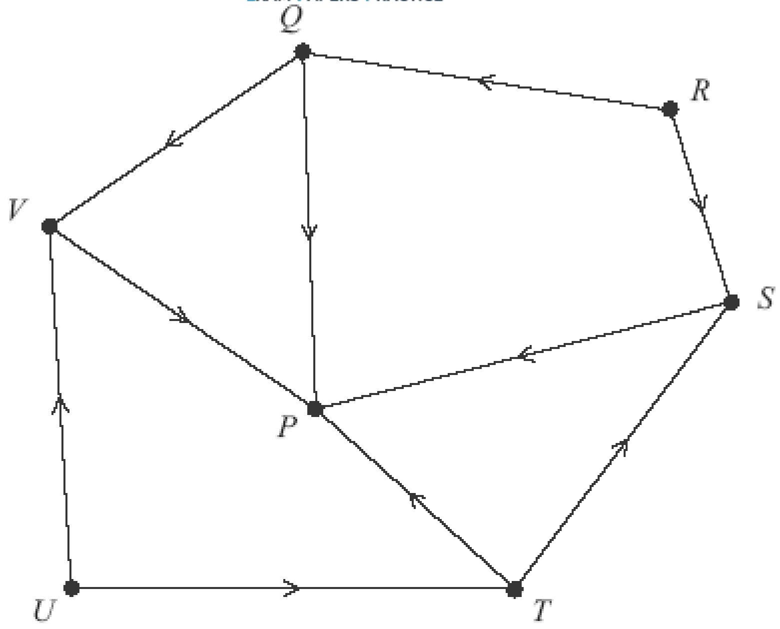
Route	Value of Flow
UTSP	
RQVP	

(6)

(ii) State the value of the maximum flow, and, on **Figure 2**, illustrate a possible flow along each edge corresponding to this maximum flow.

Figure 2

Maximum flow is ____________________ passengers per minute



(2)

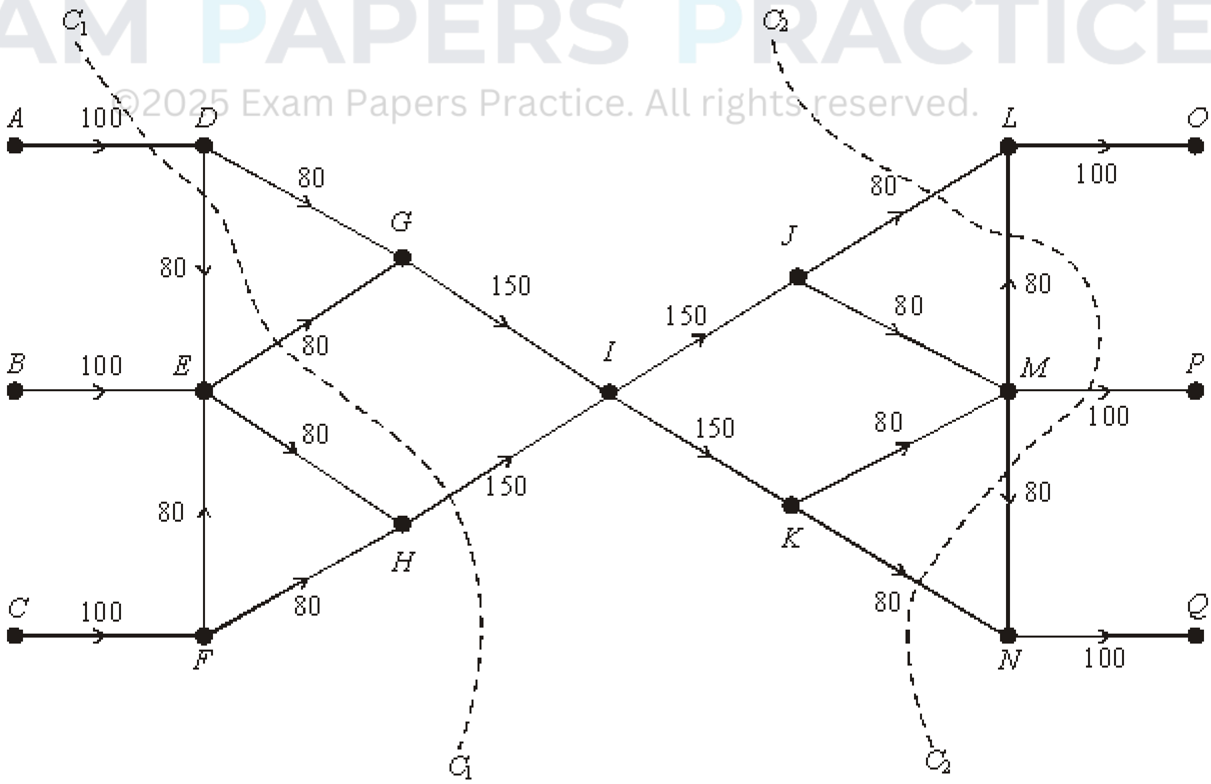
(e) On a particular day, there is an obstruction allowing no more than 50 passengers per minute to pass through vertex V. State the maximum number of passengers that can move through the network per minute on this particular day.

(2)

(Total 14 marks)

Q4.

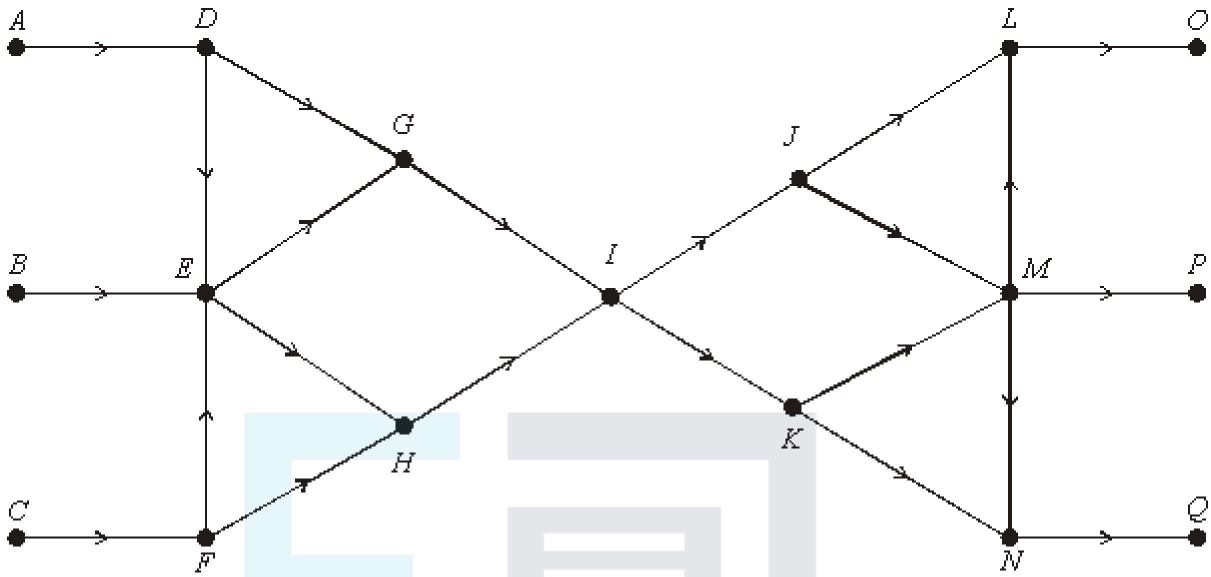
The following diagram shows the numbers of pupils who can pass along various corridors of a school building in one minute in the event of a fire drill. The classroom areas are A, B and C. The fire exits are O, P and Q.



(a) Find the values of the cuts C_1 and C_2.

(2)

(b) On the figure below, show a flow of 300 pupils per minute through the network.



(3)

(c) (i) On a certain day there is a restriction at vertex I of 120 pupils per minute.

Find the maximum flow on this day.

(1)

(ii) On another day the junction at I is clear but there is a restriction at vertex H of 120 pupils per minute.

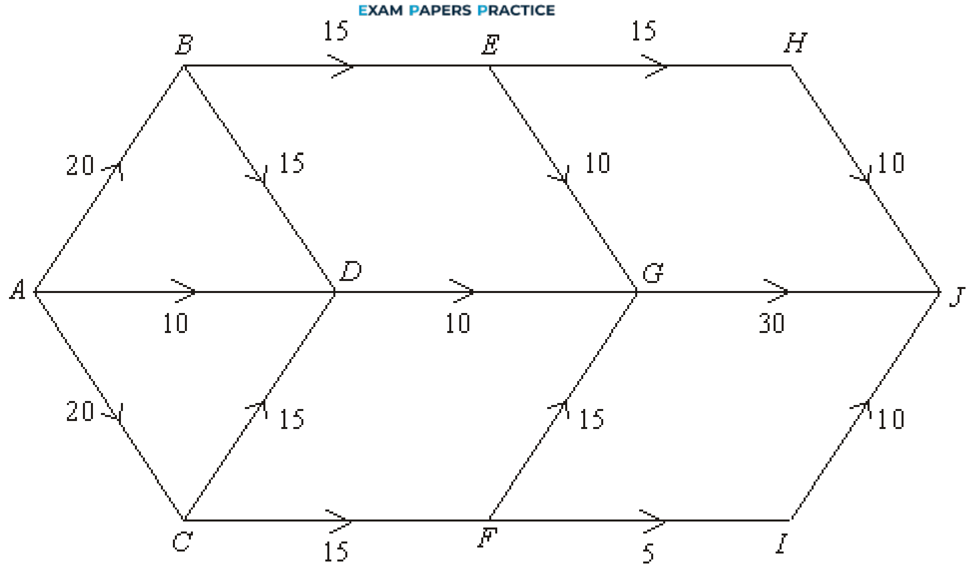
Find the maximum flow on this day.

(2)

(Total 8 marks)

Q5.

The following diagram shows the maximum number of cars that can travel along each section of a one-way system in a 5 minute interval.



(a) On the figure above, starting from a position of no flow of cars along the one-way system, use flow augmentation to find the maximum flow.

(4)

(b) Confirm that you have a maximum flow by finding a cut of the same value.

(1)

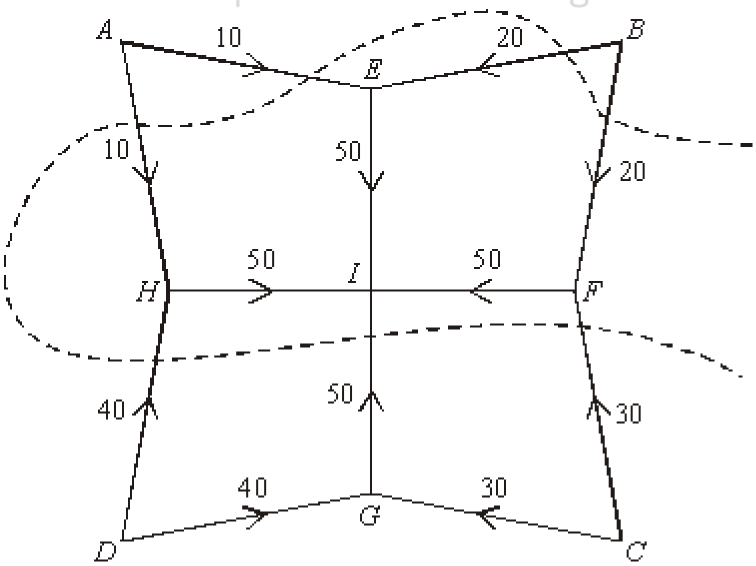
(c) A new road is to be added to the system. Determine which two vertices this new road should connect in order to make a flow of more than 50 cars, in a 5 minute interval, possible.

(2)

(Total 7 marks)

Q6.

The following diagram represents a network of pipes used in a water system in a garden. The number on each arc represents the maximum volume of water, in cubic centimetres, that can flow along the corresponding pipe in one second.



(a) State the vertices that represent the sources.

(1)

(b) State the vertex that represents the sink.

(1)

(c) Find the value of the cut X.

(2)

(d) (i) On **Figure 1**, starting from a position of zero flow, use flow augmentation to find the maximum flow.

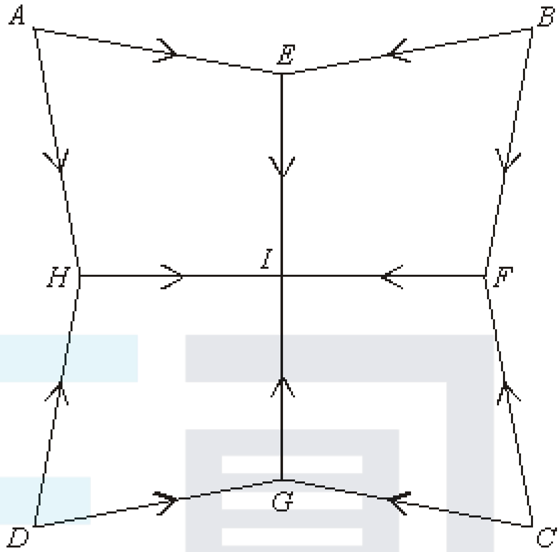


Figure 1

(4)

(ii) State why this flow is maximum.

(1)

(e) On a particular day there is a restriction at vertex E which allows a maximum flow through E of 20.

On **Figure 2**, find, by inspection, the maximum flow through the network on this day.

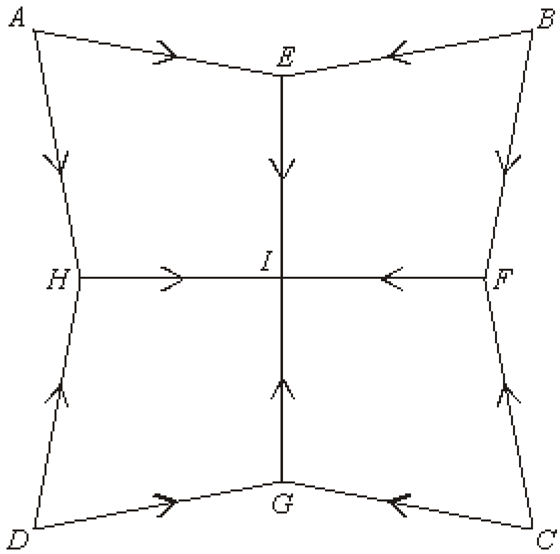
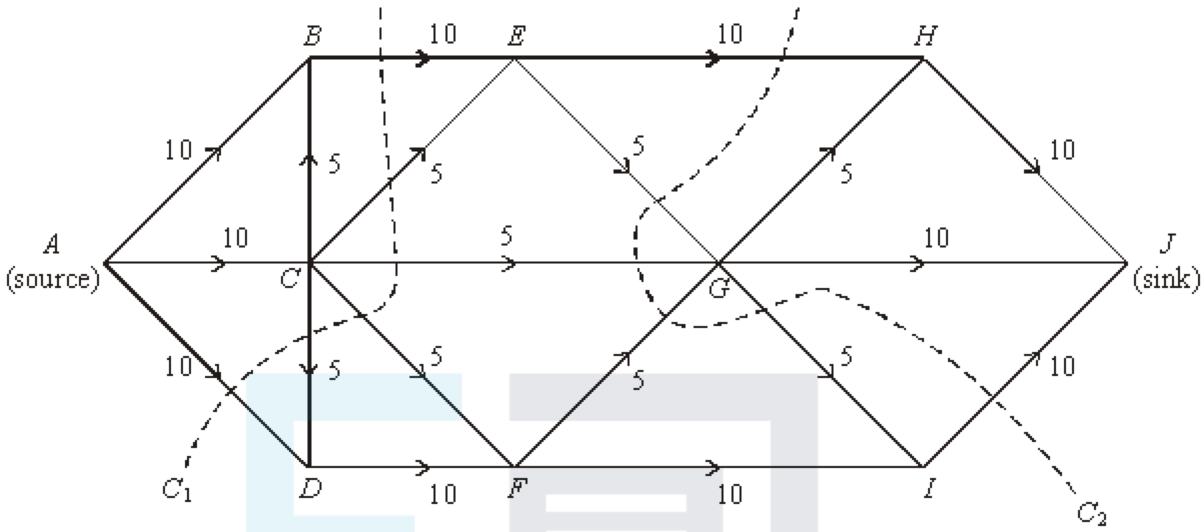


Figure 2

(2)
(Total 11 marks)

Q7.

The following diagram shows ten vertices connected by a number of arcs. The number on the arcs represent the maximum flow along each arc.



(a) Find the values of the cuts C_1 and C_2.

(2)

(b) Find, by inspection, a flow of 30 units and illustrate it on **Figure 1**.

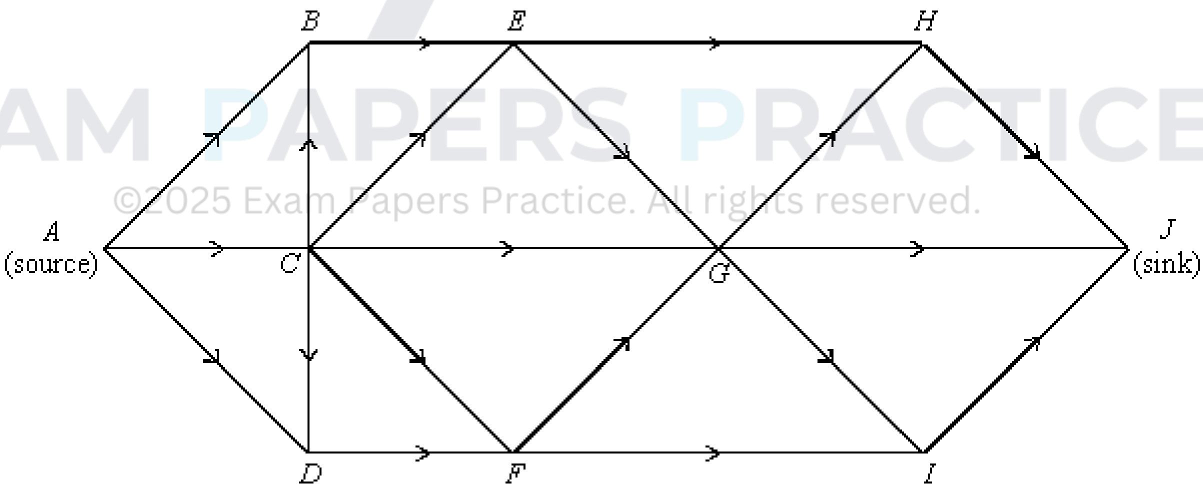


Figure 1

(2)

(c) On a certain day, the section GJ is blocked so that the network is now indicated on **Figure 2**.

Find, by inspection, the maximum flow on this day and illustrate it on **Figure 2**.

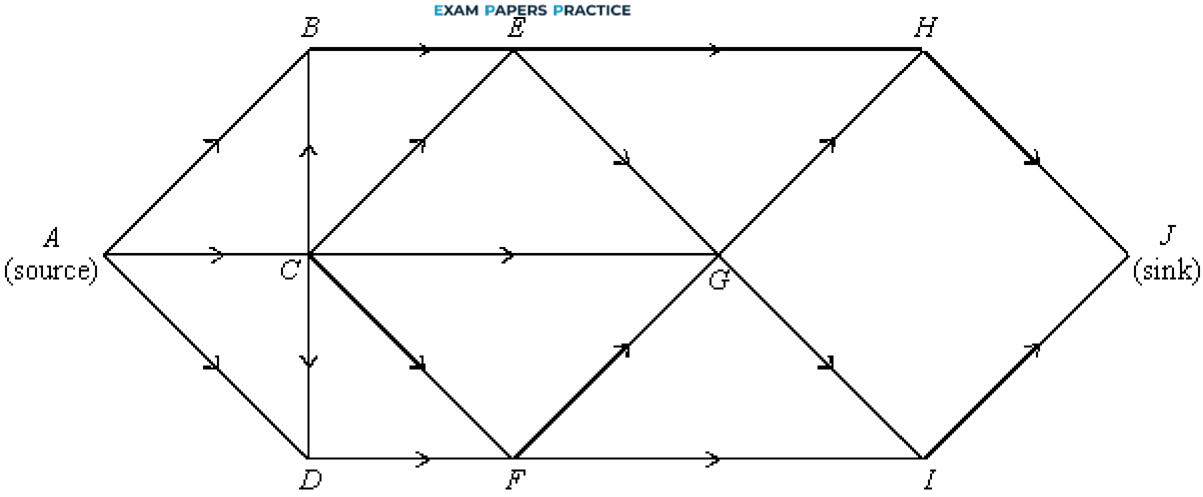


Figure 2

(2)

(d) On another day, GJ is working normally but EH is now blocked, as shown in **Figure 3**.

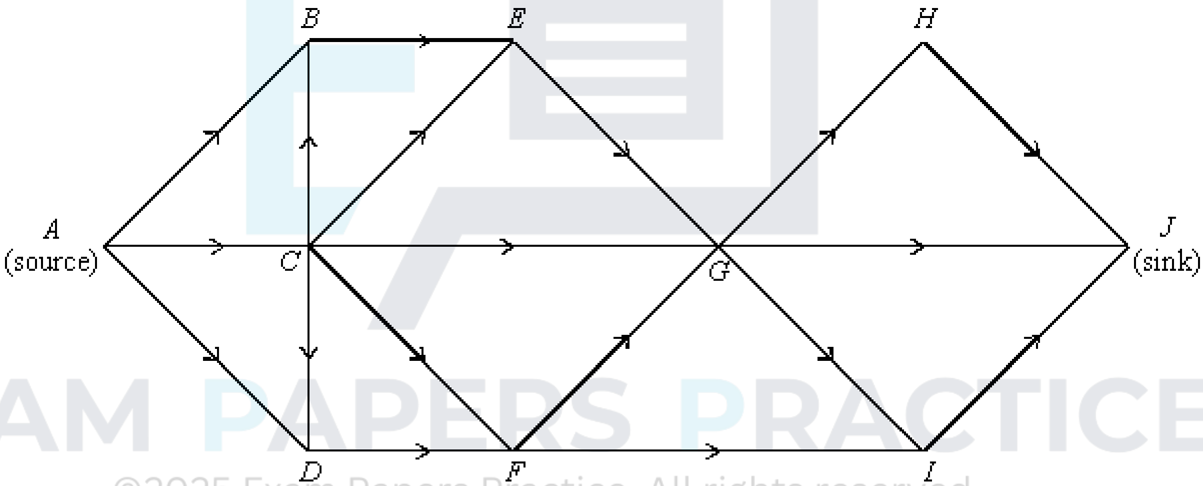


Figure 3

(i) Find, by inspection, the maximum flow on this day and illustrate it on **Figure 3**.

(2)

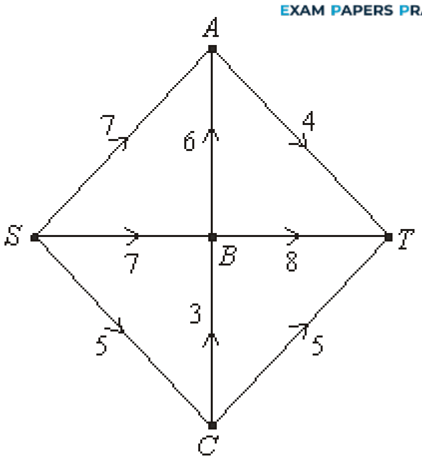
(ii) Show that the flow obtained in part (d)(i) is maximal.

(2)

(Total 10 marks)

Q8.

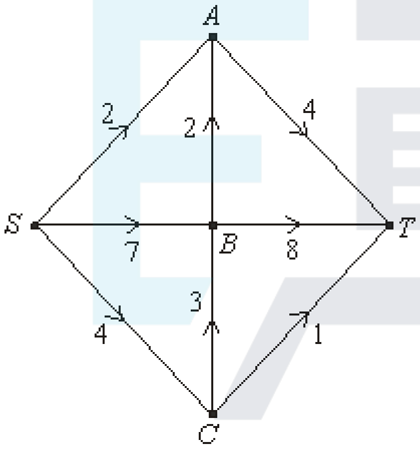
The diagram shows the capacities of the arcs of a network.



(a) Find a cut of 16 in the network above.

(2)

(b) The next diagram shows a flow of 13 from S to T in the same network.



(i) Find a flow-augmenting path from S to T which uses just one other vertex and which raises the given flow of 13 to a flow of 14.

(2)

(ii) Find a further flow-augmenting path from S to T which raises the flow from 14 to 16.

(2)

(c) How do you know that the flow of 16 is the maximum possible in this network?

(1)

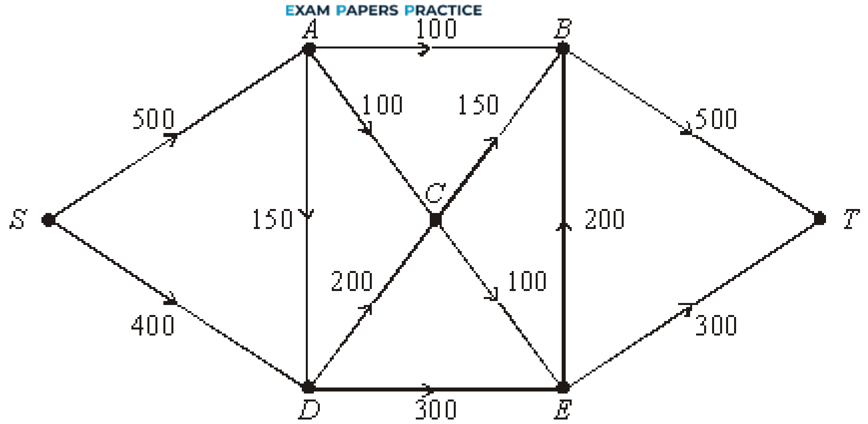
(d) The capacity of one arc of the network is to be increased in order to enable a flow of 19 from S to T. State which arc should be chosen and give a reason.

(3)

(Total 10 marks)

Q9.

The following network shows a one-way system of roads: the number on each arc is the maximum number of cars which can travel along that road in one hour.



(a) Starting from a zero flow, carry out repeated augmentations until you find a maximum flow from S to T. Show your workings on **Figure 1** and give full details of the maximum flow on **Figure 2**.

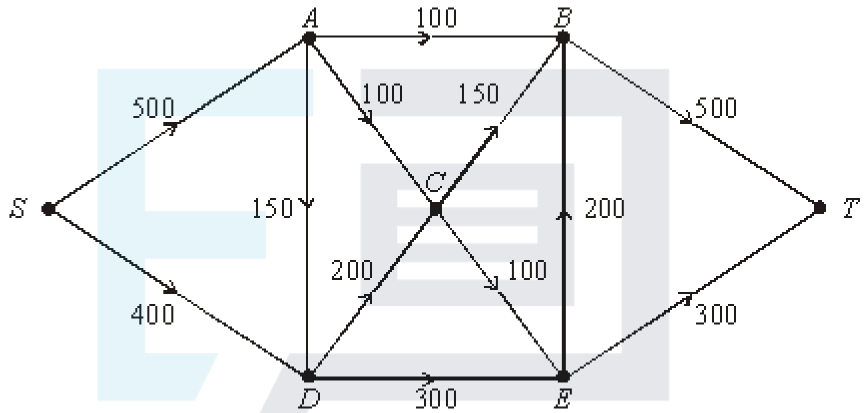


Figure 1

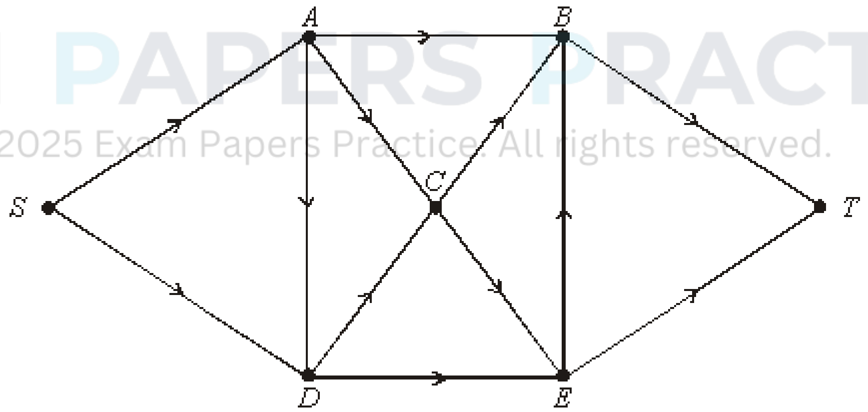


Figure 2

(6)

(b) Confirm that the flow which you found in part (a) is a maximum by finding a cut of the same value.

(2)

(c) A large underpass is to be built to take traffic directly from C to T. In the corresponding new network state how to augment your answer to part (a) in order to increase the flow by 50 cars per hour. Show, however, that no further increase is possible.

(3)
(Total 11 marks)